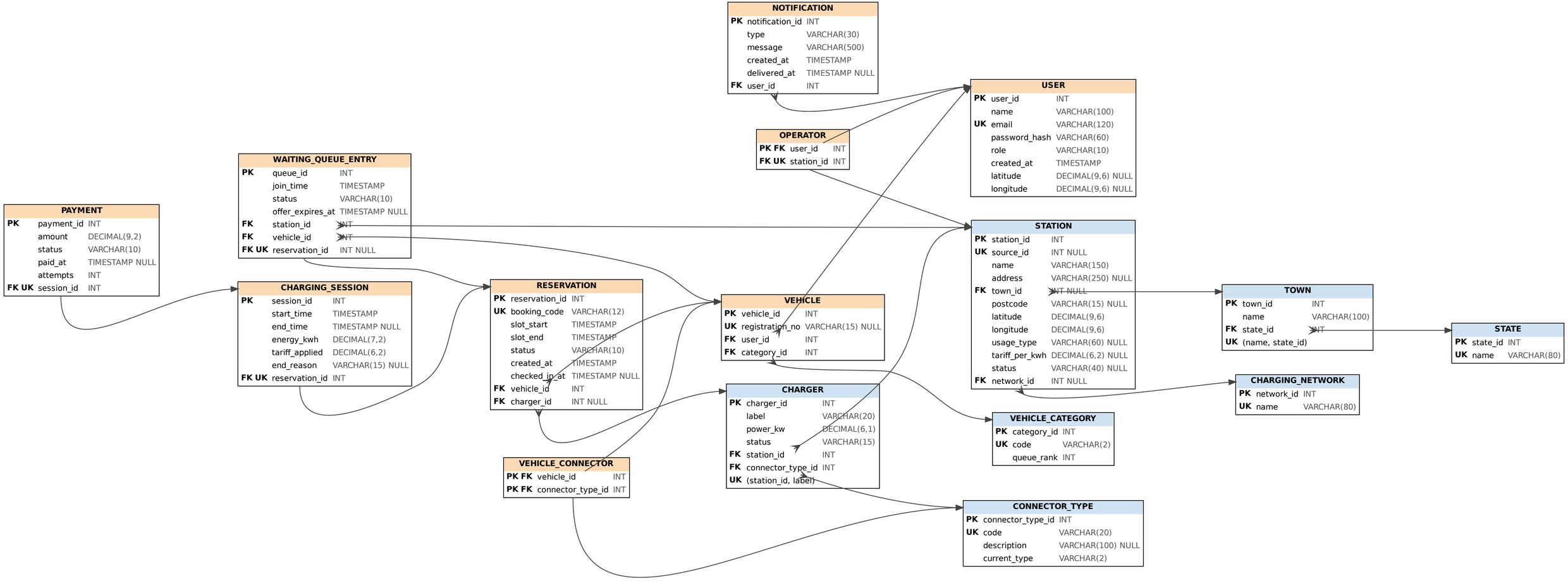
EV Charging Station Network Orchestrator — Relational Schema (FK → PK), Revised v3 (3NF)

Blue header: predefined-data table Orange header: user-data table PK primary key, FK foreign key, UK unique, NULL nullable. Arrow runs from FK column to referenced table. Crow's foot = many, plain end = one (unique FK).



Revision summary (v3): 3NF fixes

Issue	Dependency	Change in v3
CHARGING_SESSION.cost	(energy_kwh, tariff_applied) → cost	Column removed. Derived value via view v_charging_session_cost (ER: dashed attribute).
PAYMENT.amount	session_id → cost → amount	Kept on purpose as a billing snapshot (frozen at payment time). Trigger checks it equals energy_kwh × tariff_applied on insert.
STATION town / state	postcode → town → state	STATE and TOWN tables added (town → state). STATION.town, STATION.state replaced by FK town_id (nullable). postcode kept as a plain attribute. New relationships LOCATED_IN (TOWN 1:N STATION) and PART_OF (STATE 1:N TOWN).
WAITING_QUEUE_ENTRY.priority_class	vehicle_id → category_id → priority	Column removed. Derived from VEHICLE_CATEGORY via view v_queue_priority; displaced entries flagged by reservation_id IS NOT NULL.
CHARGER natural key	(station_id, label) → charger_id	UNIQUE (station_id, label) added.

Derived values (views)

```
CREATE VIEW v_charging_session_cost AS
  SELECT session_id, ROUND(energy_kwh * tariff_applied, 2) AS cost FROM charging_session;

CREATE VIEW v_queue_priority AS
  SELECT q.queue_id, vc.code AS priority_class, vc.queue_rank, (q.reservation_id IS NOT NULL) AS is_displaced
  FROM waiting_queue_entry q JOIN vehicle v ON v.vehicle_id = q.vehicle_id
  JOIN vehicle_category vc ON vc.category_id = v.category_id;
```

New tables: STATE(state_id, name), TOWN(town_id, name, state_id). Result: every determinant in every table is a candidate key, so the schema is in 3NF (and BCNF).